Yihui Wang

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Ph.D. student, ECE, University of Maryland | neuromorphic and asynchronous IC design | two tapeouts, TSMC 180/65 nm

Education

University of Maryland, College Park — **Sep 2024 – Present**
Ph.D. in Electrical Engineering, expected 2029 – GPA 3.81/4.0 – Advisor: Dr. Sahil Shah — *College Park, MD*

George Washington University — **Sep 2021 – Dec 2023**
M.S. in Electrical Engineering – GPA 3.83/4.0 — *Washington, DC*

Yanshan University — **Sep 2016 – Jun 2020**
B.Eng. in Photoelectron Information Science and Engineering — *Qinhuangdao, China*

Silicon Tapeouts

SpineLink: 32-Neuron Asynchronous Spiking-Neural-Network Chip — **Taped out Jun 2026**
TSMC 180 nm | 1 mm^2 die | Mixed-signal | Genus, Innovus, Virtuoso, Calibre

- **Architecture:** clockless tree–spine NoC of 18 bundled-data micropipeline routers, 1,152 SPI-programmed 3-bit synapses, and asynchronous SerDes links for chip-to-chip chaining.
- **Implementation:** RTL to GDS; core integrated with 32 analog LIF neurons in Virtuoso; full-chip Calibre DRC/LVS closed.
- **Post-layout: 3.8 ns and 17 pJ per router hop**, 412 fJ per spike, 1.9 µW digital static power.

Synthesizable Asynchronous AER Encoder — **Taped out May 2025**
TSMC 65 nm | 1 mm^2 die | Digital | Genus, Innovus, Virtuoso, Calibre

- **Design:** 8-event bundled-data AER encoder built from standard cells only; runs through the stock synthesis and PnR flow.
- **Silicon: 33 MEvent/s, 50 ns latency, 435 fJ/event**, 8.7 µW measured; published at IEEE MWSCAS 2026.

Research Experience

Shah Lab, University of Maryland — **Sep 2024 – Present**
Graduate Research Assistant – designer on the two tapeouts above — *College Park, MD*

- **MURI program:** contribute to the ARO MURI on hybrid artificial–living neural networks (lead: Prof. Wolfgang Losert, UMD), which seeks to identify the learning rules of biological neurons and replicate them in silicon spiking neural networks.
- **Bio-LSM readout:** spiking-neural-network decoder that classifies responses of a simulated biological neuron culture and identifies the most informative readout timescale (9.2 pJ per synaptic op; npj manuscript).
- **NoC exploration:** evaluated five network-on-chip topologies against the sparse, event-driven traffic of spiking neurons using closed-form area and latency models and synthesized routers.
- **SpineLink NoC:** proposed the resulting tree–spine NoC, with **2.9× less router area** than a 2-D mesh at 1,024 neurons and **17 vs 48 nJ per broadcast spike**; carried to tapeout in TSMC 180 nm (above).
- **Clockless methodology:** design method for standard digital flows so power scales with neural activity: micropipelines with automated timing closure; proven on 65 nm silicon, open source.
- **Current work:** a 28 nm asynchronous LIF core with local SRAM synapses, higher extendability NoC structures, and small-world topology networks to support larger-scale neuromorphic systems with far-less memory and power consumption.

ADAM Lab, George Washington University — **Sep 2023 – Sep 2024**
Graduate Research Assistant — *Washington, DC*

- **Izhikevich SNN on FPGA:** 16-bit fixed-point network with pipelined, time-multiplexed neurons (open source, 8 stars).

Projects

LIF Spiking Decoder on FPGA **2026**

Digilent Arty S7-50 (Spartan-7) | *Vivado* | *100 MHz* *Verilog, Python*

- Designed an 811-input, 10-neuron LIF decoder: Q6.4 membrane, 4-bit weights, one event per clock, 6 Mbaud UART link.
- Achieved **97.9% accuracy** (snnTorch network reference) at **9.2 pJ per synaptic operation**; 1,323 LUTs, 0 DSPs.

EEG Spike Encoder and Izhikevich SNN SoC **May 2025**

SkyWater 130 nm | *Genus, Innovus* | *mixed-signal* *Verilog*

- Designed a 12-neuron Izhikevich network in 16-bit fixed point; 21k cells, 0.28 mm^2, zero timing violations.
- **43 µW** digital power at 3 kHz; **106.5 µW** full system with analog LNA and delta-modulation encoder at 300 Hz.

Digital Design Flow **(open source, 18 GitHub stars)** **2025 – 2026**

Genus, Innovus, Xcelium | *TSMC 180 / 65 / 28 nm, SkyWater 130 nm* *Tcl, Shell*

- RTL-to-GDS Tcl flow with post-synthesis and post-route SDF simulation; used for both tapeouts and by lab members.

MIPS32 5-Stage Pipelined CPU **Nov 2023**

AMI 0.5 µm | *Design Compiler, Innovus* *Verilog*

- Forwarding, stall and branch-flush hazard handling; scan chains inserted for DFT; timing closed at **50 MHz**.

Publications

1. **Y. Wang**, S.-Y. Peng, and S. Shah, "An Ultra-Low-Power Synthesizable Asynchronous AER Encoder for Neuromorphic Edge Devices," *IEEE MWSCAS*, 2026 (accepted). arXiv:2604.05313.

2. J. A. Miller, **Y. Wang**, C.-Y. Lin, S. Shah, and W. Losert, "The Choice of Timescales in Hybrid Biological–Artificial Networks," *npj Unconventional Computing*, 2026 (under review).

3. **Y. Wang** and S. Shah, "SpineLink: A Fully Asynchronous Tree–Spine NoC with In-Network Broadcast and Convergecast for Multi-Chip Neuromorphic Systems," *IBM IEEE CAS/EDS AI Compute Symposium*, 2026 (poster, accepted).

Technical Skills

- **Design:** RTL and asynchronous design, RTL-to-GDS, STA/SDC, DFT, DRC/LVS sign-off, FPGA, silicon test
- **EDA tools:** Cadence Genus, Innovus, Voltus, Xcelium, Virtuoso, Spectre; Synopsys Design Compiler; Calibre; Vivado
- **Languages:** Verilog/SystemVerilog, Verilog-A, Tcl, SKILL, Python (PyTorch, snnTorch), C/C++, Shell, MATLAB